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Department of Climate Change, Energy, the Environment and Water

Nature Repair Market Mapping Guidelines

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Nature Repair Market Branch

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Acknowledgement of Country

We acknowledge the Traditional Owners of Country throughout Australia and recognise their continuing connection to land, waters and culture. We pay our respects to their Elders past and present.

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Introduction

The *Nature Repair Act 2023* (the Act) came into effect on 15 December 2023 establishing a framework for a world-first legislated, national, voluntary biodiversity market. The Act provides legislated rules to support transparency and integrity and to foster collaborative efforts to address environmental decline.

The Nature Repair Market Scheme is a government initiative that incentivises actions to restore and protect the environment. It encourages nature positive land management practices that deliver improved biodiversity outcomes. The scheme establishes a marketplace where individuals and organisations can undertake nature repair projects to generate a tradable certificate.

This document applies to biodiversity projects carried out under the requirements of the Act. It establishes requirements for the preparation of geospatial data and maps where required under the Rules or methods.

This document will be made available on the Department's website and linked to from the Clean Energy Regulator's website. For further guidance and information about the scheme requirements refer to the Clean Energy Regulator and DCCEEW websites.

1. When do these guidelines apply?

This document is intended as an incorporated document under the Rules and methods, where specified in these legislative instruments. The Rules and methods detail how and when this document is to be applied. This may include, but is not limited to, providing geospatial data and maps for a project area when:

- applying to register or vary a biodiversity project, including the development of a project plan;
- determining the biodiversity outcome for a biodiversity project, and measuring progress in achieving the outcome;
- complying with biodiversity project reporting, record-keeping and monitoring obligations; or
- applying for a Biodiversity Certificate.

Reference should be made to the Rules and any applicable method to determine when to apply this document.

These guidelines align with the processes involved in developing a biodiversity project using PLANR.

2. Glossary and acronyms

The tables below provide an explanation of key terms and acronyms used in this document.

Where a term used in this document is defined in the Nature Repair Act 2023 or applicable legislative instruments, that definition applies. The Nature Repair Act 2023 and legislative instruments made under that Act are available at www.legislation.gov.au.

Term	Explanation
Activity area	An area or areas within a project area where activities are to be carried out for the purpose of the biodiversity project (whether there are specified requirements to carry out the activity or specified requirements not to carry out the activity).
Attribute	Non-spatial information about a geographic feature in a GIS, usually stored in a table and linked to the feature. For example, an activity area might have an activity area id, condition type etc.
Coordinate system	Provides a reference for measuring horizontal and vertical distances on a map. It is usually defined by a map projection, spheroid reference, datum and other parameters such as standard parallels, central meridian and possible shifts in the x and y directions.
Datum	A set of parameters and control points used to accurately define the three-dimensional shape of the Earth.
Exclusion area	Areas within the project area that biodiversity activities are not or cannot be carried out in.
Feature	With respect to a GIS dataset, a single record that has both geometry and attribute data.
Geometry	The measures and properties of points, lines, and surfaces. In a GIS, geometry is used to represent the spatial component of geographic features.
Geographic Information System (GIS)	Means a geographic information system designed to capture, store, manipulate, analyse, manage, and present all types of geographical data.
Geospatial data	Refers to information that describes the locations and characteristics of features on the Earth's surface. It includes both spatial data, which specifies the geographic location of features (such as coordinates or boundaries), and attribute data, which provides details about these features. This data can be represented in various interchangeable digital GIS formats and is essential for analysing patterns and relationships in a geographic context.
Interchangeable digital GIS formats (data file)	Standard means of encoding geographical information into a computer file including geometry, attributes and map projections. The format used for the original file must be able to be changed into other file formats without loss of information contained in the original file.

Term	Explanation
Line segment	A straight line bounded by two end points, with each end point having distinct map coordinates.
Map	A map is a visual representation of an area, displaying geographic features, boundaries, and important locations.
Map projection	A mathematical model or set of mathematical equations that describe the relationship between the same points on three-dimensional and two-dimensional surfaces and transform the spatial relationships of features on the Earth's three-dimensional surface onto a two-dimensional surface. Distortions in area, shape, direction and distance are minimised.
Metadata	Structured descriptive data that provides information about another set of data. Metadata can include, but is not limited to, data title, data source, and accuracy.
Multipart polygon	A collection of polygons, used for gathering a group of polygons into one geometry representing common attributes. Sometimes referred to as multi-part polygon or feature.
Platform for Land and Nature Repair (PLANR)	The Platform for Land and Nature Repair provides landholders with information about the attributes of their property and plan biodiversity projects.
Orthorectify	Removal of distortions in an image so the new image has a consistent scale.
Polygon	A closed shape bounded by a single connected polyline that does not cross, such that the map coordinates of the end point of the polyline are the same as the map coordinates of the start of the polyline. Polygons can define the boundaries of a project area, activity area, or exclusion area. Subject to the applicable method, polygons can contain holes which exclude their enclosed area from the polygon.
Polyline	A continuous line composed of one or more connected line segments.
Project area polygon	A polygon defining the external boundary of a project area. Project areas which are located in more than one place will require a set of project area polygons to define their external boundaries.
Stratification	The subdivision of the land within a project area according to biophysical characteristics and management regimes, or by other characteristics as allowed or required within a specific method.
Sub-area	As defined in relevant method.
the Act	Refers to the *Nature Repair Act 2023.*
the Rules	Refers to the *Nature Repair Rules 2024*.

Acronym	Meaning
ANZLIC	Spatial Information Council of Australia and New Zealand
BAI	Biodiversity assessment instrument

Acronym	Meaning
FSDF	Foundation Spatial Data Framework
GDA	Geocentric Datum of Australia
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GPS	Global Positioning Systems
ICSM	Intergovernmental Committee on Surveying and Mapping
LiDAR	Light Detection and Ranging
MGA	Map Grid of Australia
PLANR	Platform for Land And Nature Repair
UAV	Unmanned Aerial Vehicles
UTM	Universal Transverse Mercator

3. Mapping requirements

3.1 Identify the Project Area

A project area is the area within which a biodiversity project will be carried out in. All biodiversity projects must have a project area defined as part of the application process for registering a biodiversity project.

The project area is subject to specific requirements and limitations outlined in the Act, Rules and the applicable method. This includes whether the project area may be based on land, in water, or a combination of both, and minimum size of project area requirements.

A project area may take several forms, including:

- A single polygon (Figure 1 left).
- A multipart polygon, consisting of areas separated by a distance (Figure 1 right)

Figure 1: Examples of project areas that have simple (left) and complex multipart (right) boundaries

A proponent should consider the ownership or legal arrangements that apply to the project area. This would include being aware of land title(s) (where applicable), ownership arrangements or regulatory approvals relevant to the proposed project area. A project area will generally cover all area on which a proponent has the right to carry out a project.

The outer boundaries of a project area may encompass the entire land title(s) area, even if only part is involved in biodiversity activities (Figure 2).

Defining boundaries of the project area must be established using the tools, data sources and standards defined in these guidelines.

Failure to correctly identify the project area and compliance with necessary legal requirements related to the project area may affect project registration, require a project variation to be undertaken or require proponents to reapply for project registration.

Figure 2: Project area outer boundary (white dashed line) in relation to property (cadastre) boundaries.

3.2 Stratification of a Project Area

Stratification is the process of dividing a project area into distinct areas based on the management activities to be implemented and the underlying biophysical characteristics within the project area. The applicable method for a biodiversity project will outline how the project area should be stratified.

The stratification process divides the project area into activity areas. Any other areas within the project area are designated as exclusion areas.

Stratifying the project area ensures that the unique ecological properties of each part of the area are considered during planning a biodiversity project and that the management activities are appropriate.

Each section of the project area will be grouped according to shared characteristics (Figure 3a). This will form the basis for creating activity areas.

3.3 Identify Activity Areas and Sub-areas

Activity areas are the areas where project activities, eligible with a method, will be carried out. The applicable method will determine whether activity areas are required for a biodiversity project and provide guidance on how to define. Methods may contain additional requirements for activity areas, including specifying the minimum size, dimensions of activity areas, and conditions regarding eligibility.

Activity areas are created by aggregating areas based on similar biophysical characteristics (Figure 3b). This ensures that project activities (e.g. replanting, habitat improvements or threat management) are implemented consistently across an area.

Each activity area will be attributed according to the requirements set out in a method. At a minimum an activity area must be unique (Table 1).

Table 1: Activity area minimum attribute requirements

Field Name	Description	Data Type	Example	Notes
ActivityID	Unique identifier for each activity area	Text (Alphanumeric)	AA1	Must be unique across all activity areas.
Area_ha	The area of the activity in hectares	Float	12.35	Calculated for each activity area.

Methods may require that activity areas are divided further into sub-areas based on a reference ecosystem. Methods may contain additional requirements for sub-areas, including specifying the minimum size, dimensions and biophysical characteristics.

Each sub-area will be attributed according to the requirements set out in the method. At a minimum a sub-area must be unique (Table 2)

Table 2: Sub-area minimum attribute requirements

Field Name	Description	Data Type	Example	Notes
SubArea_ID	Unique identifier for each sub-area	Text (Alphanumeric)	SA1	Must be unique across all activity areas.
ActivityID	Identifier for the activity area for which the sub-area relates to	Text (Alphanumeric)	AA1	Relates sub-area to its activity area
Area_ha	The area of the activity in hectares	Float	12.35	Calculated for each activity area.

3.4 Exclusion areas

Exclusion areas are areas within an activity area that biodiversity project activities are not or cannot be carried out in. This may be because the proponent chooses to not include the area in the project, or because it is not possible to carry out a particular type of biodiversity project in certain areas. Examples of exclusions may include a large rock outcrop, existing remnant vegetation, utility easement or an access road, cultural heritage sites, or an area that doesn't meet the eligibility requirements set out in a method.

Exclusion areas must be identified in an activity area for:

- features greater than five metres in width; or
- areas less than five metres in width that collectively make up more than five per cent of the activity area

For example, in an activity area of one hectare (10,000 m^2), exclusion areas should include all areas where the project mechanism does not apply, as long as these areas are more than 500 m^2.

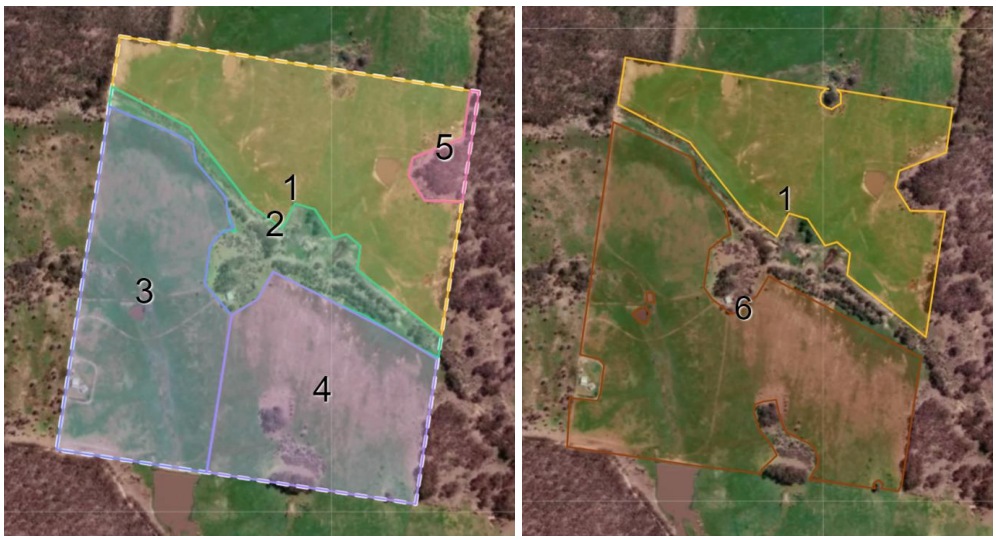

Figure 3a: Example of a stratification of a project area into biophysical characteristics **3b:** Activity areas that are eligible as a biodiversity project.

3.5 Varying project and activity areas

Project area

Project proponents must apply to the Regulator if they want to vary a project area. Varying a project area includes changing the project area boundaries, irrespective of whether that change results in an increase, decrease or no change to the size of the project area. The requirements for varying a project area are established in the Act and the Rules.

Activity area

Where permitted by the applicable method, the project proponent may need to revise an activity area to ensure the project remains able to achieve the biodiversity outcome for the registered biodiversity project. Examples include when:

- Disturbance events occur
- Biodiversity project activities are no longer uniform across the activity area
- Site characteristics are no longer uniform
- Activity areas are no longer viable to achieve the required biodiversity outcome

3.6 Sampling plots

Sampling plots are defined spatial regions used to systematically collect site-specific data about activity areas.

Each sampling plot should have a clearly defined centreline running through its length, with start and end points accurately mapped using GPS to ensure positional accuracy. Methods will outline the number, dimensions, and placement of these sampling plots.

Each sampling plot must have a unique identifier and must be relatable back to an activity area (Table 3).

Table 3: Sampling plot minimum attribute requirements

Field Name	Description	Data Type	Example	Notes
Plot_ID	Unique identifier for each sampling plot within an activity area	Text (Alphanumeric)	PL001	Must be unique for each sampling plot.
ActivityID	Identifier for the related activity area	Text (Alphanumeric)	AA1	Relates sampling plot to its activity area

3.7 Creating a map

A map is a visual representation of an area, displaying geographic features, boundaries, and important locations. Maps are generally provided within a document, such as a project plan, to assist readers of that document. A map is different to geospatial data provided in a digital format.

When submitting a map to the Regulator, it is important to include specific map elements to ensure clarity, accuracy, and compliance with the requirements (Figure 4).

The following are key features that must be included on any mapping product:

Boundary: The boundary of the mapped area must be clearly marked. It should encompass the entire area where biodiversity activities will be carried out. Dataset should be included as overlays on the map.

Map Scale: The map must include a scale that indicates the relative size of the area. This helps anyone reading the map to understand the size and extent of the area shown.

Coordinates: Include geographical coordinates (latitude and longitude) or grid references for the corners (vertices) of the mapped area. This ensures that the areas can be precisely located on the ground.

Legend: A legend must be included to explain the symbols, lines, and colours used on the map. For example, you can use different colours to differentiate between the project area and activity areas.

North Arrow: Include a north arrow to orient the map correctly. This helps users understand the direction of the area being mapped.

Title and Date: The map should have a clear title, identifying what the map is representing, and the date it was created or last updated.

Land Features: Any significant natural or man-made features (e.g. rivers, roads, or infrastructure) within or around the mapped area should be included for reference.

Data Source and Projection: State the data source used to create the map (e.g., satellite imagery, property boundaries from a cadastral database) and the map projection (e.g. GDA94, Map Grid of Australia). This ensures the map can be reproduced and verified if necessary.

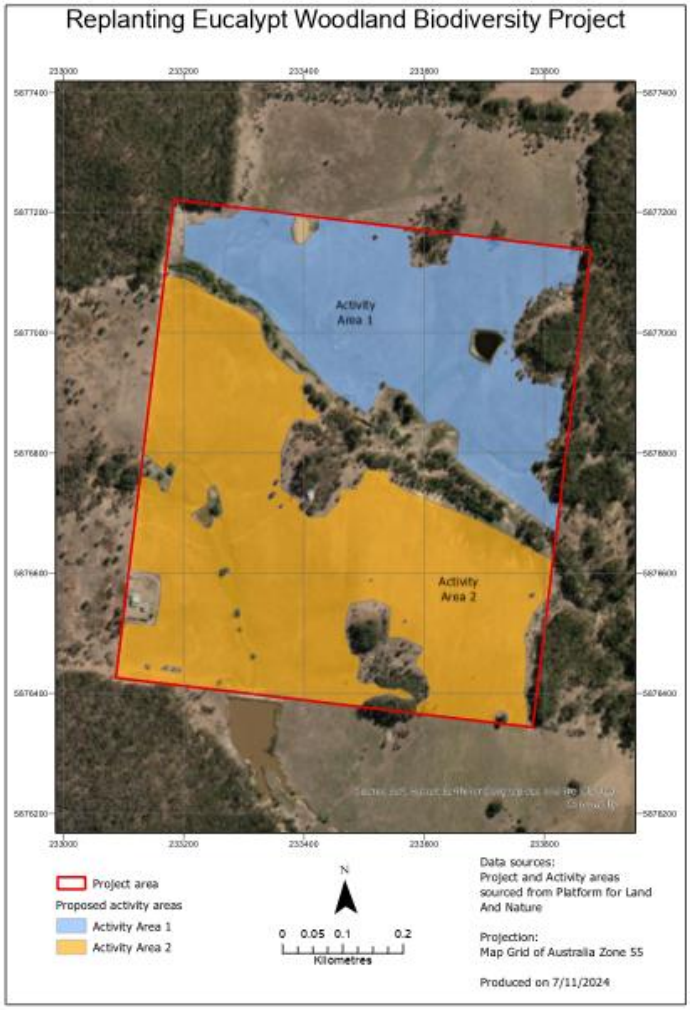

Figure 4: Example of a map and key elements

4. Geospatial information requirements

4.1 Geospatial capture

A project proponent may use any of the following sources of data to delineate boundaries:
Technologies:

- **Global Positioning Systems (GPS) / Global Navigation Satellite System (GNSS)**: Offers precise field coordinates for accurate boundary mapping and field sampling applicable with a method.
- **Unmanned Aerial Vehicles (UAVs)/Drones:** Increasingly used for high-resolution, low-altitude aerial surveys and mapping specific to biodiversity projects.

- **Aerial photographs**: Useful for visualising large areas and identifying landscape features.
- **Satellite imagery**: Ideal for covering large areas remotely, providing up-to-date imagery
- **Light Detection and Ranging (LiDAR)**: Provides highly accurate elevation and terrain data, useful for generating Digital Elevation Models (DEMs) and understanding landscape features relevant to project areas.

Data Sources:

- **Cadastral databases:** Official property boundaries collected by jurisdiction ensure legal accuracy in mapping.
- **Publicly available environmental datasets**: Public datasets include both national and state resources. These datasets can include information on vegetation types, topography, land use, land cover, first nations, water resources, and more.
- **Site assessment datasets**: datasets generated by a suitably qualified person in preparation of an application to the Regulator.
- **PLANR:** access to essential datasets required for biodiversity projects under the Act.

A method may also specify additional data sources, approaches, accuracy or resolution requirements to be complied with.

Records used to identify and delineate geospatial boundaries may be subject to reporting and recordkeeping requirements under the method.

4.2 Accuracy

For all projects, the minimum requirement for spatial data is a horizontal accuracy of at least 10 metres (m) at 95 per cent (%) threshold in accordance with the Intergovernmental Committee on Surveying and Mapping (ICSM) Australian Map and Spatial Data Horizontal Accuracy Standard 2009. Evidence of this data accuracy must be included within accompanying metadata.

Some methods may place additional requirements for the horizontal and vertical accuracy of spatial data being supplied.

4.3 Fitness for purpose

Prior to using an existing dataset, project proponents - or a suitably qualified person - must evaluate the appropriateness of the dataset for its intended use to ensure that it is fit for purpose. This evaluation should consider:

- **Reputable Source**: The dataset should come from a reliable and reputable source that is preferably publicly available and supported by comprehensive metadata to ensure transparency and traceability.
- **Age and Currency**:
    - Some methods may require the use of satellite imagery or aerial photographs sourced within a specified time period
- **Dataset Scale**: Ensure that the dataset scale matches the required level of detail. Large scale datasets (e.g. 1:10,000) should be used for detailed site assessments and boundary delineation as it provides greater precision. Small-scale datasets (e.g. 1:100,000) are better suited to broader regional planning but lack the necessary detail for project-specific areas.

- **Imagery Resolution**: It is important to use high-resolution imagery that captures detailed features. Use imagery with a resolution of 1 metre per pixel or finer. This resolution allows for accurate identification of small features such as fence lines and individual trees.
- **Positional Accuracy**: Evaluate how well the dataset's spatial data corresponds to actual real-world locations. Higher accuracy is especially important for tasks like boundary delineation or site-specific assessments.
- **Classification and Generalisation**: Consider the classification, aggregation, or generalisation systems used within the dataset. For example, some datasets may smooth or simplify data in ways that affect its suitability for precise tasks.
- **Integrity and Coverage:** Ensure the dataset covers the entire project area and is free from significant gaps or inconsistencies that could affect the accuracy of the analysis.

4.4 Coordinate System

Geospatial data supplied to the Regulator must use the Geocentric Datum of Australia (GDA94 or GDA 2020) datum.

The recommended projected coordinate system is the Map Grid of Australia (MGA94 or MGA2020), based on the Universal Transverse Mercator (UTM) projection.

4.5 Data exchange and formats

Project proponents must supply spatial data in an interchangeable digital format which is consistent with the instructions outlined in these mapping guidelines.

Data must be supplied in a zipped file containing a single dataset of one of the common GIS file types (e.g. Esri geodatabase, Esri Shapefile, GeoJSON, Mapinfo or GeoTiff). The preferred format is a shapefile or geodatabase.

When creating a zipped file of shapefile, include all of the shapefile components in addition to the 4 essential components (that is, .dbf, .prj, .shp and .shx, and also .cpg, .sbn, .sbx and .shp.xml).

4.6 Geospatial datasets and attributes

Dataset requirements

The geospatial dataset requirement by method (Incorporated documents and resources) lists the geospatial datasets that are required for each method and application type (i.e. project registration application, project area variation application and certificate application), in accordance with the Act, Rules, applicable method and these guidelines.

Each geospatial dataset must be uploaded separately in a zip file. There is an upper size limit of 500MB per upload. The Regulator will assess datasets to ensure they meet the requirements set out in these guidelines. Geometries are checked to identify geometry problems using the Check Geometry (Data Management) function.

Attribute requirements

Project proponents must ensure that all geospatial files contain all attributes and conform to the data type requirements. Geospatial datasets supplied in support of applications will be subject to validation checks to ensure they meet these requirements.

Where applicable, geospatial data must include the following:

- A unique identifier for each feature being described (e.g. ActivityID).
    - This identifier must:
        - Be unique for each feature within the dataset
        - Consist of a combination of alphanumeric characters, spaces, underscores, en dashes, periods, commas, and parentheses
        - Be relatable between different datasets. For example, the ActivityID in the activity area dataset must be relatable to the ActivityID in the sampling plot dataset when referencing.
        - Be used consistently across filenames and dataset submissions.

Project proponents may include other attributes in their geospatial dataset to assist administration of their project or supply additional information to the Regulator.

Additional information about each feature may be required in reports submitted to the Regulator and must reference the unique identifier provided in the geospatial dataset.

4.7 Data ownership and usage

Project proponents must ensure that they have the necessary rights, for example, a data licence for primary mapping products, to use spatial data for the project. Proponents must also be able to grant the Regulator or its representative permission to use a project's spatial data for the purpose of administering the Nature Repair Market.

In line with disclosure arrangements and mechanisms within the Act, data will be made publicly available via the Biodiversity Market Register. Sensitive data will be managed in accordance with the Regulator's Privacy Policy to ensure appropriate protection of personal and sensitive information.

4.8 General instructions

Geospatial mapping information must meet all the following requirements:

- describe the coordinate system and map projections used
- be submitted electronically and use one of the commonly used interchangeable digital GIS file formats;
- contain file information on:
    - the feature geometry (that is, the map coordinates of the line segments that make up the project area polygons);
    - a positional index of the feature geometry (that is, the location within the file of the start of the data for each geometry);
    - attribute information; and
    - the coordinate system and map projection used.